Tutorial – 03

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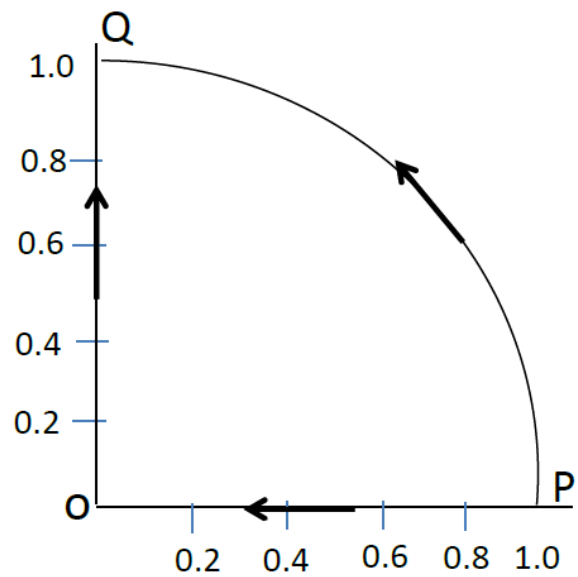
1. Evaluate $\iint \boldsymbol{A}.\hat{\boldsymbol{n}}dS$ where $\boldsymbol{A} = z\hat{\boldsymbol{\imath}} + x\hat{\boldsymbol{\jmath}} - 3y^2z\hat{\boldsymbol{k}}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant $z = 0$ and $z = 5$.

2. A particle is moved under the influence of a force $\boldsymbol{F}(x, y) = -ax\hat{\boldsymbol{\imath}} - by\hat{\boldsymbol{\jmath}}$ from point $P(1,0)$ to pint $Q(0,1)$ in two ways: (i) along the axes ($P \to O \to Q$), (ii) along the quarter circle ($P \to Q$). Here 'a' and 'b' are constants with appropriate units



(a) Calculate the work done by the force in both the cases

(b) Do the two results match? What does it tell you about the nature of force?

(c) Can $\boldsymbol{F}(x, y)$ be expressed as the gradient of a scalar function? If yes, try to obtain it.

(d) Evaluate curl of $\boldsymbol{F}(x, y)$. Is the answer consistent with above results?

3. Calculate the Laplacian of the following:

(a) $g(x, y, z) = \frac{1}{12}x^4 - ln\frac{y}{z}$

(b) $\vec{\boldsymbol{V}}(x, y, z) = e^{-xyz}\hat{\boldsymbol{\imath}} + z^2\hat{\boldsymbol{\jmath}} + xz\widehat{\boldsymbol{k}}$